Workshop Title: AI Part 2 – From Algorithms to Ethics

Time: 60 Minutes

Goal: Understand recommendation systems, demystify LLMs, and explore ethics through play.

1. Section 1: The "Invisible Hand" – Recommendation Systems (15 Mins)

Explain that AI isn't just a chatbot; it's the "invisible hand" that decides what we see.

- **The Jane Example (Collaborative Filtering):** * **Scenario:** Jane watches *The Godfather*.
    - **The Math:** AI doesn't just look at the movie; it looks at **Bob**. If Bob liked *The Godfather* and also loved *Goodfellas*, the AI assumes Jane and Bob are "tastes-mates." It recommends *Goodfellas* to Jane because someone *like her* liked it.
- **The YouTube Loop (Neural Networks):** * YouTube uses 80 billion signals daily. It doesn't just care what you watched; it cares about **Watch Time** (did you finish it?), **CTR** (did you click it?), and **Context** (are you on your phone at 2 AM or on a TV at 6 PM?).

Specifically, it's the "fuel" for a type of AI called a **Deep Neural Network**.

Here is how you can explain this to your group:

1. What is a "Signal"?

Think of a signal as a tiny breadcrumb of behavior. Every time someone interacts with YouTube, they leave a trail.

- **Explicit Signals:** What you *tell* YouTube (Likes, Dislikes, Subscribes, clicking "Not Interested").
- **Implicit Signals (The "Secret Sauce"):** What you *show* YouTube without saying a word.
    - **Watch Time:** Did you watch 10 seconds or 10 minutes?
    - **Retention:** Did you skip the intro? Did you re-watch a specific part (the "Dopamine Hit" signal)?
    - **Context:** What time is it? Are you on a phone (likely looking for a quick Short) or a Smart TV (likely settling in for a long documentary)?
    - **The "Next" Move:** Did you leave the app after this video, or did it make you stay for three more?

2. What are the "80 Billion Signals"?

This number represents the **total volume of these breadcrumbs** that YouTube's system processes every single day.

- It means the system isn't just looking at *your* 50 clicks; it's looking at the trillions of interactions from 2 billion users and narrowing them down into 80 billion high-quality data points to find patterns.

3. How does this relate to Neural Networks?

In your first session, you defined Neural Networks. Here is how to bridge it:

- **The Problem:** You have 800 million videos. You have a user (Jane). Which 10 videos do you show her?
- **The Neural Network Solution:** YouTube uses a **Two-Stage Neural Network**:
    1. **Candidate Generation:** A "coarse" neural network looks at your history and narrows 800 million videos down to a few thousand.
    2. **Ranking:** A "fine-tuned" neural network takes those thousands of videos and scores them based on those **80 billion signals**. It predicts: *"Given Jane's history, there is a 75% chance she will watch this specific video for at least 5 minutes."*

When you open a brand-new account, YouTube doesn't just show a blank screen. It uses **"Non-Personalized Prior Signals"** to fill your feed.

The 3 Ways YouTube Solves the "Cold Start":

- **1. Demographics & Context (The "General Vibe"):** Even without a history, the AI knows three things immediately: **Your Location, Your Device, and the Time of Day.**
    - *Example:* If you open the app in **Matawan, NJ** on a Saturday morning from an iPhone, it might show you trending news in the Tri-State area, popular Saturday morning cartoons, or viral "Shorts" currently trending among other iPhone users in your region.
- **2. Popularity & Freshness (The "Billboard" Effect):** The AI defaults to what is globally or regionally "safe." It looks for videos with a high **"Velocity"** (gaining views very fast right now). You'll see the MrBeasts of the world or major news events because, statistically, a random human is likely to click them.
- **3. The "Exploration" Phase (Testing You):** This is where the AI gets smart. It will intentionally throw **diverse "bait"** at you. It might show one cooking video, one sports highlight, and one tech review.
    - **The Second-by-Second Signal:** If you hover your mouse over the cooking video for 2 seconds but click the tech review, the "Candidate Generation" layer *instantly* updates. Your next refresh will be 50% tech.

A great piece of "trivia" for your group: As of late 2023, if you turn off your **Watch History** and have no previous data, YouTube's homepage will actually appear **blank** for many users, showing only the search bar.

The Lesson for the Group: This proves that the AI is "starved" without your data. It would rather show you *nothing* than show you "bad" guesses that might make you leave the app.

- **Activity**

  https://artsandculture.google.com/experiment/say-what-you-see/jwG3m7wQShZngw?hl=en

- **The Connection: Explain that just as they are trying to guess the AI's "internal logic," the YouTube algorithm is trying to guess *their* internal logic using the signals they provide.**

Group activity --

Display these choices

The "10 Video Choice Menu" (Use this for all three people)

Group selects top 2 videos to display that you think she might watch

1. **Tech: 2026 Smartphone Review**
2. **Health: 10-Minute Morning Stretch**
3. **Gaming: High-Stakes Esports Highlights**
4. **Cooking: Quick 5-Minute Breakfast Ideas**
5. **Comedy: "Try Not To Laugh" Cat Videos**
6. **Finance: Stock Market Opening Bell Report**
7. **Relaxation: Deep Sleep Rain Sounds**
8. **Education: "How a Nuclear Reactor Works"**
9. **DIY: Repairing a Leaky Faucet**
10. **News: Local Weather & Traffic Update**

Persona 1: The Early Riser

Demographics: Female, 70s, Retired.

Time of Day: 7:00 AM

- **The "Right" AI Answers: #2 (Morning Stretch) and #10 (Weather & Traffic).**
- **The Logic: AI assumes a senior morning routine often involves physical wellness and checking the day's conditions. It bypasses gaming and high-energy comedy because historical data for this demographic at this hour shows a preference for "settling into the day."**

Persona 2: The Night Owl

Demographics: Male, 20s, Student/Gamer.

Time of Day: 2:00 AM

The "Right" AI Answers: #3 (Esports Highlights) and #7 (Sleep Rain Sounds).

The Logic: This is a "Split Decision." The AI sees it's 2 AM. It will test if the user is in "Active Mode" (Gaming) or "Shutdown Mode" (Sleep Sounds). It won't show #6 (Finance) because the stock market is closed and that demographic rarely checks it at 2 AM.

Persona 3: The Commuter

Demographics: Male, 30s, Professional.

Time of Day: 8:15 AM (Standing on a train platform)

he "Right" AI Answers: #6 (Finance) and #4 (5-Minute Breakfast).

The Logic: The AI recognizes the "Commute Window." It prioritizes short, information-dense content that can be consumed on a mobile device without much audio (or with headphones). It might also show #1 (Tech Review) as "transit entertainment." It avoids #9 (DIY Repair) because you can't fix a faucet on a train.

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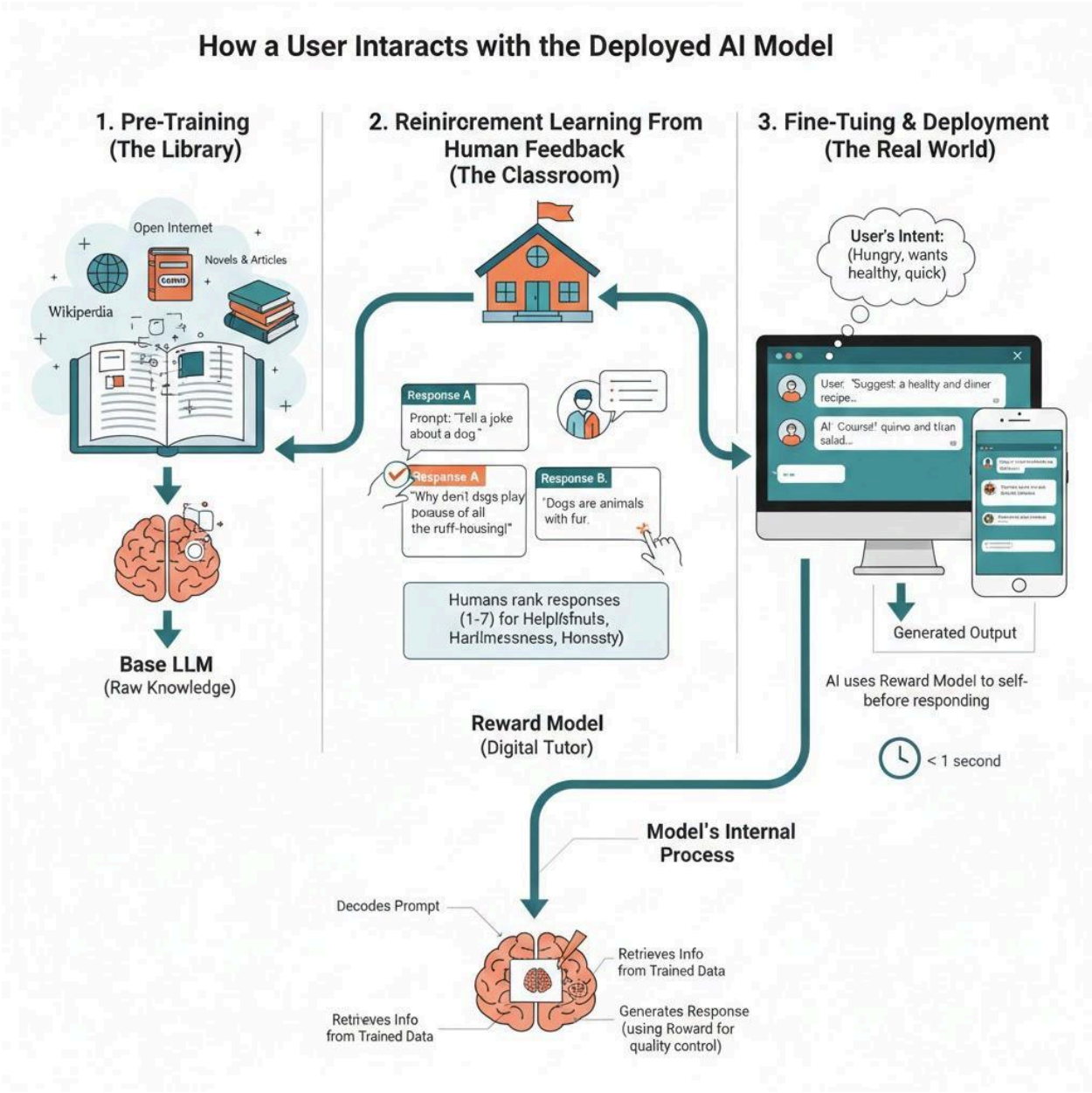
2. Section 2: ChatGPT – The Super-Predictor (10 Mins)

Demystify the "magic" of the LLM.

- **What is a Model?** Compare it to a **digital chef** who has read every recipe book in the world but has never actually tasted food.
- **The Learning Process:**
    - **Pre-training:** It read the "open internet" (Common Crawl, Wikipedia, books).

    - **Is it learning now?** No. It’s "frozen" in time (its knowledge cutoff). It doesn't "learn" from your chat in real-time to update its global brain, but it *remembers* the context of your current conversation.
    - **The "Right" Examples:** It uses **RLHF (Reinforcement Learning from Human Feedback)**. Humans literally sat there and ranked its answers, saying "This is a good, helpful answer" and "This is a hallucination/mean answer."
- but it is a massive, real-world operation. Companies like OpenAI and Google hire thousands of people (often called **Data Labelers** or **AI Tutors**) to act as the "moral compass" and "quality control" for the AI.
- The humans did this work **months ago** to build a **"Reward Model."** * Think of the Reward Model as a "Digital Human." The AI runs its answers through this "Digital Human" first to see if it would get a high score. If the Digital Human says "That looks like a hallucination," the AI tries again before showing you the result.



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3. Section 3: The Ethics "Trolley Problem" (15 Mins)

Use your Institute of AI Management notes to keep it grounded.

- **The "Garbage In, Garbage Out" Rule:** Mention the **COMPAS judicial system bias**—if the data is biased against specific groups (like the black defendants in the COMPAS tool), the AI will be too.
- **Practical Industry Brief:**
    - **Hiring:** Resume scanners might favor "people like us" (Sample Bias).
    - **Healthcare:** Algorithms trained only on lighter skin tones fail to detect issues on darker skin.
- **Interactive Ethics Activity (The "Hallucination Hunt"):**
    - Have everyone ask their AI: *"Tell me three facts about [Your Local School District/Matawan History] that are definitely true."* * Ask them to find one small detail the AI got wrong. This teaches **Transparency** and **Accountability**.

Goup exercises ->

1. Do you write yoru wedding vows? Authenticity vs assistance. If memories are real, does the editor matter? The “effort” signal - in relationships, is the work of writing the gift, or the words themselves? Betrayal factor → if your spouse found out 10 years later it was AI, would they be hurt?
2. Best Man’s speech - use AI to generate a speech, what if you provide the input like personal storis, his interests, how he met his fiance’, and just used AI to structure it?
3. Valentines Day card - does handwriting make it yours? Intent vs execution - you want to bve sweet, is that intention good enough?
4. Homework help - do you do your son’s homework for him? Learning vs completion - is the goal to finish or understand the lesson? Is using calculator wrong? Does using AI at age 10 present the mental musles from developing?

What is ethical? Combination of this:

Transparency: If you use it, do you tell people? (The "Betrayal" test).
Accountability: If the AI makes a mistake in your wedding vows (e.g., calls your wife the wrong name), who is responsible? You are.
Fairness: Is it "fair" to the other students if Gabriel uses AI and they don't?
The "Paul's Principle": Suggest that **AI is a co-pilot, not the pilot.** It can help you find the words, but you must provide the heart.

The Paul Ireifej AI Transparency Mandate

> *"If it’s not 100% human, the human 100% needs to say so."*

In the future, every digital communication (Social Posts, Texts, Emails, Videos) should carry a mandatory **"AI Nutrition Label."** We don't hide ingredients in our food; we shouldn't hide them in our thoughts.

The 4 Categories of Disclosure

We need to move past "Is this AI?" and ask **"How much AI is this?"**

Label	The Level of Use	Example Use Case
[AI-Synthetic]	The AI created the entire thing from a prompt.	A generic LinkedIn post or a generated "stock" photo.
[AI-Co-Pilot]	A human had the idea and the "soul," but the AI organized, edited, or polished the words.	A Best Man's speech where the AI turned your memories into a script.
[AI-Enhanced]	A real human photo or video that was "cleaned up" or "upscaled" by AI.	Removing a stranger from a family photo or fixing "grainy" video.
[Human-Only]	Zero AI involvement. Raw, unedited, and authentic.	A handwritten note or a live-recorded voice message.

The AI Alert Warning System

Just like a weather alert on your phone, our digital interfaces should have a **Visual Code** to warn us about the "authenticity" of what we are seeing.

1. The "Logic" Light (Text)

- **Green Check:** Content has been verified against reputable sources (matches your class notes on .gov/.edu data).
- **Yellow Warning:** Content contains "unverifiable" claims (Potential Hallucination Alert).
- **Red Flash:** Content is high-probability misinformation or generated to mimic a specific person without a "Co-Pilot" tag.

2. The "Digital Fingerprint" (Media)

- **The Invisible Watermark:** Every image generated must have a "Provenance Marker" baked into the code. If you try to post it without the **[AI-Synthetic]** tag, the platform automatically adds a "Ghost Stamp" over the image that says: *Created by AI.*

Closing Discussion Point for the Group

> *"Transparency isn't about banning AI; it's about preserving the value of the human. If everything is tagged, then a 'Human-Only' message becomes the most valuable thing you can send."*

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4. Section 4: The "Prompt-Off" Competition (20 Mins)

This is the heart of the fun. You are the judge.

Challenge 1: The Creative/Silly (The "Impossible Invention")

- **Goal:** Use AI to describe or "draw" (if using DALL-E/ImageFX) the most useless but hilarious invention.
- *Example Prompt:* "Write a sales pitch for a solar-powered flashlight for people who only walk at night."

Challenge 2: The Humorous (The "Roast My Career")

- **Goal:** Get the AI to write a 30-second stand-up comedy bit about being a [Participant's Job Title].
- *Example:* "Write a joke about the struggle of being a middle school principal in 2026."

Challenge 3: The Useful (The "Life Hack")

- **Goal:** Generate a prompt that solves a real problem in under 10 seconds.
- *Example:* "I have a half-empty jar of pickles, a block of cheddar, and an onion. Give me a snack recipe that doesn't taste like regret."

The "Teachable" Tool: AI vs. Human (Google's "Say What You See")

Since you've used Teachable Machine, try **Google's "Say What You See"**.

- **How it works:** It shows an AI-generated image, and you have to guess the prompt to recreate it.
- **Why it fits:** It perfectly demonstrates how "semantic meaning" works and why clear instructions (Prompt Engineering) are necessary to overcome the AI "black box."

Would you like me to draft the specific "judging criteria" for your Prompt-Off to keep it feeling professional yet lighthearted?